

Cleaning Response-Time Distributions: Removing the Help-Timeout Censoring Artifact

MULTIPLY project — fluency-practice first-attempt response times

Method note: C1 baseline reconstruction with smooth forward redistribution

This note documents the procedure used to clean first-attempt response-time (RT) distributions of the fluency-practice data before distributional modelling. The raw per-fact RT densities contain sharp artificial spikes at the Leitner help-timeout times; the procedure removes those spikes, reallocates their mass forward in time, and returns a smooth curve, while leaving the genuine pre-timeout data untouched.

1. The artifact

Pooled per-fact RT densities show a smooth retrieval peak (around 2–4 s) followed by sharp, narrow spikes at fixed times — 10, 15, 20 and 30 s. These spikes are not a property of retrieval; they are an instrument artifact produced by the practice software's help mechanism, and if left in place they distort every summary statistic (mean, tail, skewness) and corrupt any distribution fitted to the data.

2. Why it happens: Leitner help timeouts as censoring

Each fluency card is scheduled by a Leitner box system. A card sits in a box that fixes a **time before help**: when the pupil has not answered by that time, a help prompt fires and the recorded first-attempt duration piles up at the timeout. The timeout depends on the box:

Box	Review interval	Time before help (timeout)
A	1 day	30 s
B	3 days	20 s
C	5 days	15 s
D	10 days	10 s
E	25 days	10 s
Memory	150 days	10 s

Statistically the spike is **right-censoring**: a pupil piled at the timeout did not retrieve in time, so their true RT is only known to be greater than the timeout. Because a single fact is practised by pupils sitting in different boxes, its pooled curve inherits a spike at every box's timeout (10, 15, 20, 30 s). A warehouse check confirmed each box is effectively capped at its own timeout (max durations $\approx 10.3 / 15.3 / 20.5 / 30.5$ s), so within a box there is essentially no data beyond the cap.

3. Design goals

The cleaning is required to satisfy four properties simultaneously:

- **Causality.** Data before a timeout is real and must never be modified. A spike at 10 s may only affect the curve at and after 10 s.
- **Mass conservation (no deletion).** The censored pupils are not discarded; their mass is reallocated forward in time, since they would have answered later.
- **Smoothness (C1).** The cleaned density must be continuous with a continuous first derivative — no kink at either edge of a former spike.
- **Non-negativity.** The result must remain a valid (non-negative) density.

4. The algorithm

Work per fact, on the box-pooled binned density (bin width $\Delta = 0.5$ s). Let $f_{\text{raw}}(t)$ be the raw density at bin centre t .

Step 1 — flag the contaminated bins.

Mark bins within a short window after each timeout T as contaminated:

$$S = \{ t : T \leq t \leq T + w, \text{ for some } T \in \{10, 15, 20, 30\} \}$$

The window half-width w (parameter SPIKE_WIN, default 1 s) captures the pile-up. Crucially the window starts at T , so bins before the timeout are never flagged and remain anchors.

Step 2 — fit a C1 baseline through the reliable bins.

Fit a shape-preserving cubic Hermite interpolant (PCHIP) through the non-contaminated bins and evaluate it everywhere:

$$b(t) = \text{PCHIP} \{ (t, f_{\text{raw}}(t)) : t \notin S \}$$

PCHIP is continuous with a continuous first derivative (C1) and does not overshoot, so it cannot produce negative densities. Because it passes exactly through every anchor, $b(t) = f_{\text{raw}}(t)$ at all reliable bins — in particular the entire pre-timeout region (including the retrieval peak) is reproduced exactly. Across each spike, $b(t)$ simply continues the local trend, giving the smooth join at the start of the spike.

Step 3 — measure the excess (censored) mass.

For each timeout T , the excess piled in the spike above the baseline is

$$M_T = \sum_{t \in S_T} (f_{\text{raw}}(t) - b(t)) \cdot \Delta \text{ (clipped at 0)}$$

This separates the bin into its natural level (kept at b , representing pupils who genuinely answered near T) and the censored excess M_T (to be moved forward).

Step 4 — redistribute forward through a smooth C1 ramp.

The excess is added back to the tail through a ramp that vanishes, with zero slope, at T :

$$r_T(t) = 1 - \exp(-((t - T)/\lambda)^2) \text{ for } t > T, \quad r_T(t) = 0 \text{ otherwise}$$

Near T , $r_T(t) \approx ((t - T)/\lambda)^2$, so both $r_T(T) = 0$ and $r_T'(T) = 0$. The cleaned density is

$$f_{\text{clean}}(t) = b(t) + \sum_T M_T \cdot [b(t) r_T(t)] / \int b(u) r_T(u) du$$

Because the added term uses a smooth ramp rather than a hard on/off region, the mass eases in from zero: there is no switch-on edge and therefore no kink at the *end* of the spike either. The parameter λ (LAM, default 2 s) sets how far past the timeout the mass is reallocated.

5. Parameters

Parameter	Default	Role
Δ (BINW)	0.5 s	Histogram bin width of the RT density.
w (SPIKE_WIN)	1.0 s	Window after each timeout used to measure the spike excess. Raise if a spike leaks past it.
λ (LAM)	2.0 s	Width of the smooth ramp; how far beyond the timeout the censored mass is reallocated. Larger pushes mass further out.
Timeouts	10,15,20,30 s	The Leitner help times; fixed by the box design.

6. Properties guaranteed by construction

- **Pre-timeout data is exact.** $b(t) = f_{\text{raw}}(t)$ at every reliable bin and the ramps vanish for $t \leq T$, so nothing left of a timeout is altered.
- **Mass is conserved.** The baseline removes exactly ΣM_T at the spikes; each ramp adds back exactly M_T . Total probability is unchanged.
- **C1 at both spike edges.** $b(t)$ is C1, and each ramp joins with zero value and zero slope at T , so f_{clean} has a continuous first derivative throughout — no kink entering or leaving a spike.
- **Valid density.** PCHIP does not overshoot and the ramp is non-negative, so $f_{\text{clean}}(t) \geq 0$.

7. Limitations

- **Reallocation shape is assumed.** The forward mass follows the smooth baseline shape and the ramp width λ ; the true landing times of censored pupils are unobserved, so this is a modelling choice, not a measurement.
- **Highest timeout.** Mass censored beyond box A's 30 s cap has little room to be reallocated (few bins exist past 30 s); this fraction is small but genuinely unidentified.
- **Excess/level split.** Deciding how much of a spike bin is 'natural level' (kept) versus 'censored excess' (moved) depends on the baseline interpolation across the window and on w .
- **Box composition.** Pooling boxes mixes mastery levels (box correlates with how well the fact is known); the reallocated tail therefore reflects the pupils who answered late, who tend to be less fluent.

8. Output and downstream use

The procedure returns a cleaned per-fact binned distribution (`df_clean: operation, rt_bin, n`) on the same schema as the raw export, with pre-timeout counts identical to the raw data. It is written to `queries/rt_distribution_table6_clean.csv` and is the input for the log-normality checks (parabola / Q-Q / skew-kurtosis) and any subsequent distribution fitting, now free of the timeout spikes.